Today:

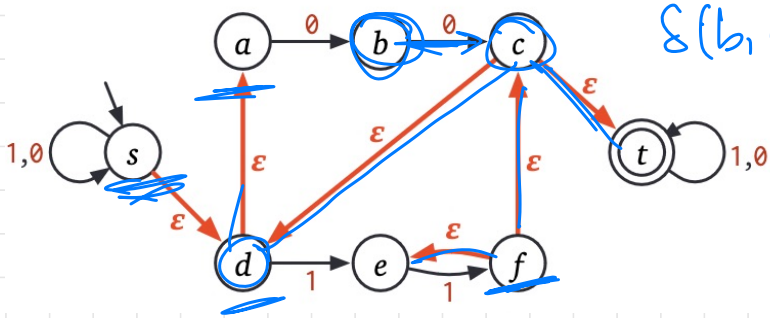
- generic transformations of NFAs/DFAs

  NFA $\longrightarrow$ DFA

- call the set of languages accepted by some DFA automatic

  set of regular langs = set of automatic langs

ε-transitions



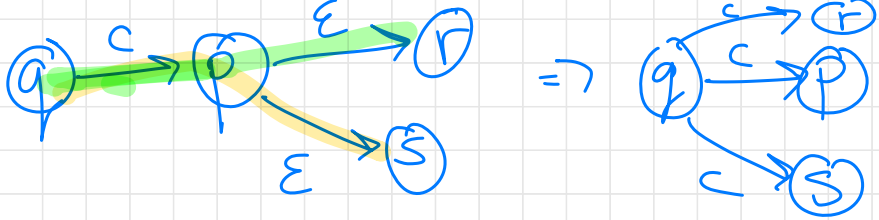
$\delta(b, 0) = c$

M

$S = S$

ε-reach(q): all states reachable from q by a sequence of ε-trans.

ε-reach$(f) = \{e, c, t, d, a, \underline{f}\}$



To convert NFA $M = (Q, S, A, \delta)$ with ε-trans. to equivalent NFA M' without ε-trans:

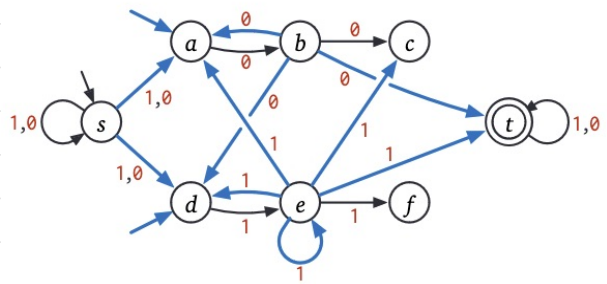
$Q' = Q$

$s' = \varepsilon\text{-reach}(s)$ $\{s, d, a\}$

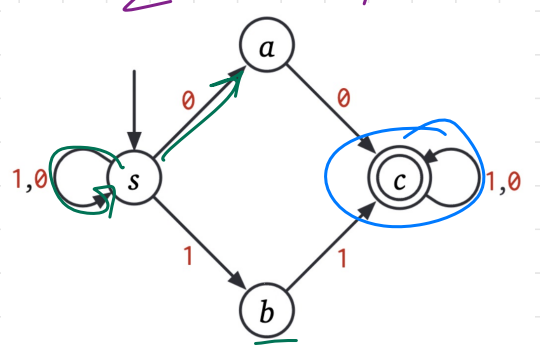
$A' = A$

$\delta'(q, a) = \varepsilon\text{-reach}(\underline{\delta(q, a)})$

$\delta'(b, 0) = \{t, d, a, \underline{c}\}$



$\Sigma = \{0, 1, 2\}$



NFA → DFA

↓

(Q, s, A, δ)

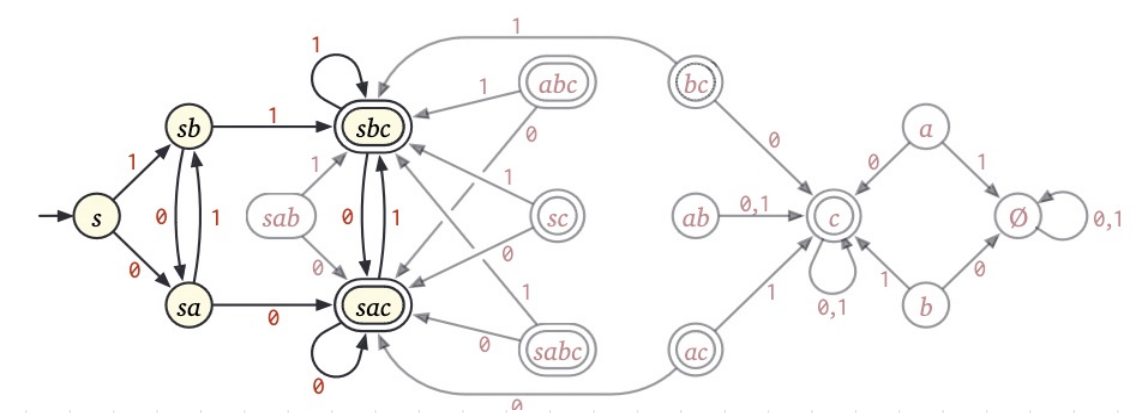
$w = 0110$

$\{s\} \xrightarrow{0} \{s, a\} \xrightarrow{1} \{s, b\} \xrightarrow{1} \{s, b, c\} \xrightarrow{0} \{s, a, c\}$

DFA: (Q', s', A', δ')

$|\text{all possible subsets of } Q| = 2^{|Q|}$

$\mathcal{P}(Q)$



DFA:

Subset construction

$Q' = \mathcal{P}(Q)$

$s' = \{s\}$

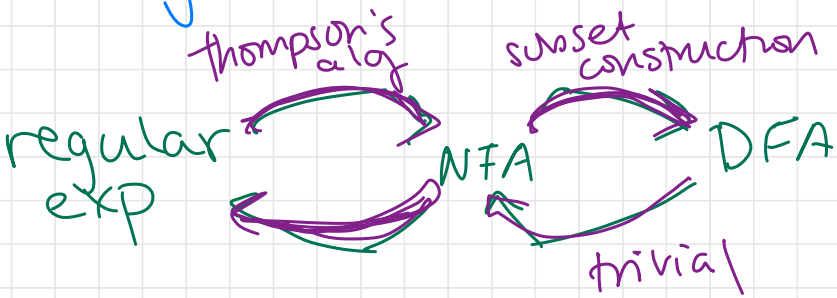
$A' = \{P \subseteq Q : A \cap P \neq \emptyset\}$

$\delta'(q', a) = \bigcup_{q \in q'} \delta(q, a)$

Kleene's Theorem (1950)

regular = automatic

Pf

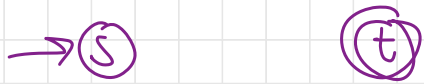


ken

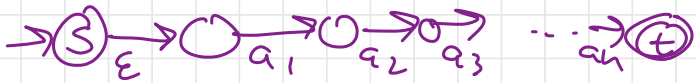
Thompson's Alg (1968)

recall def. of regular expr R:

$R = \emptyset$

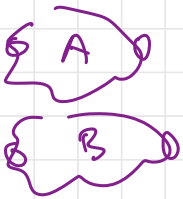


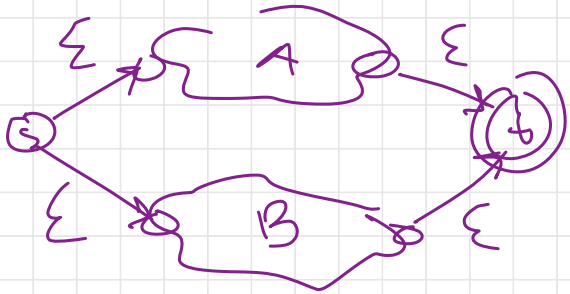
$R = w = a_1 a_2 a_3 \cdots a_n$



$R = A + B$

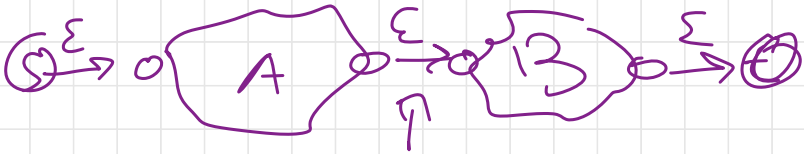
reg exps A, B





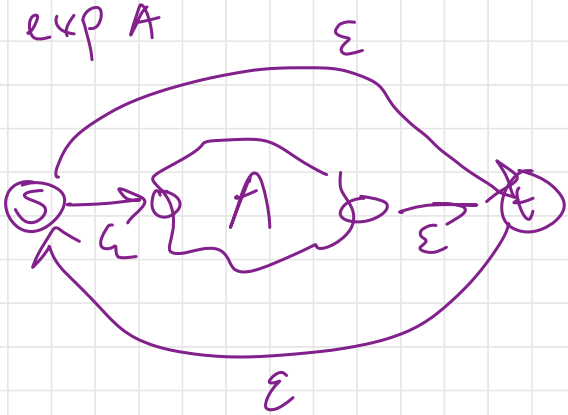
$R = A \cdot B$

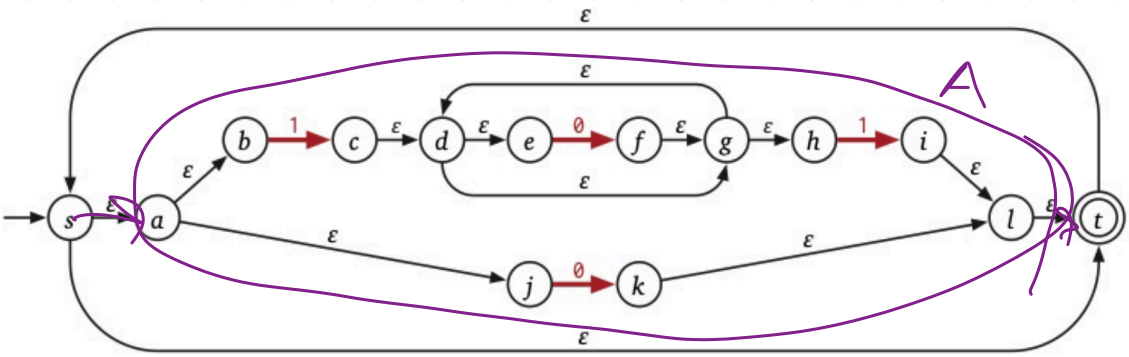
reg exps A, B



$R = A^*$

reg exp A





$$R = (10^*1 + 0)^* = (A)^*$$

Friday:

prove closure rules:

for all reg. langs L,

transform(L) is also regular

e.g.: $*$

reverse